Problem-Solution Fit

Date	5 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]	5. AVAILABLE SOLUTIONS PROS & CONS [AS]
• Farmers • Small and Medium-scale Farmers • Agricultural Students • Agricultural Officers	• Limited budget • Limited internet access • Lack of technical knowledge • Limited access to agricultural experts • Basic smartphones or computers	**Available Solutions:** • Traditional farming advice • Agricultural experts • Basic farming websites **Pros:** • Easy to access • Experienced guidance **Cons:** • Time-consuming • Not personalized • Less accurate • May not consider current soil conditions

2. PROBLEMS / PAINS
+ ITS FREQUENCY **[PR]**

- Difficulty choosing the right crop (Very High)
- Low crop yield due to poor decisions (High)
- Lack of soil analysis and guidance (High)
- Unpredictable weather affecting farming (Medium)

9. PROBLEM ROOT / CAUSE [RC]

- Lack of scientific crop recommendation
- Poor understanding of soil nutrients
- Limited access to real-time agricultural information
- Traditional farming methods
- Changing weather conditions

7. BEHAVIOR
+ ITS INTENSITY **[BE]**

- Farmers depend on experience or local advice (High)
- Use online resources occasionally (Medium)
- Want quick and accurate recommendations (High)

2. TRIGGERS TO ACT [TR]

- Low crop production
- Financial losses
- Need for better farming decisions
- Desire to improve crop yield

4. EMOTIONS
BEFORE / AFTER **[EM]**

- Confused
- Worried
- Uncertain

10. YOUR SOLUTION [SL]

OptiCrop – Smart Agricultural Production Optimization Engine

An AI-powered web application that analyzes soil nutrients (N, P, K), pH, temperature, humidity, and rainfall using a Machine Learning model to recommend the most suitable crop. The system helps farmers make informed decisions, improve crop yield, reduce financial losses, and promote sustainable agricultur

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- OptiCrop Web Application
- Agricultural Websites
- Mobile Internet

OFFLINE

- Agriculture Officers
- Farmer Meetings
- Local Farming Communities